

Causal State Binding: An Intervention-Based Evaluation Framework for Language Agents

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Abstract

Language agents retain memories and goals to guide later actions, but task success alone does not identify whether those records are used. We operationalize causal state binding through interventions on exposed agent state, combining targeted removal with irrelevant perturbation, restoration and alternative information access. Across 57,816 finite-action records from seven corpora, structured information supports consistent action responses; a separate four-model open-weight study shows advantages over removal, scrambling and distribution-matched controls. In tool-execution experiments, agents write their own state before intervention. Across 160 tasks per core configuration (80 memory and 80 goal tasks), memory removal reduces the success proportion by 0.750 in Sol (95% interval [0.656, 0.834]) and 0.541 in DeepSeek ([0.453, 0.622]); goal removal reduces it by 0.366 and 0.259. Restoration recovers performance, and redundant facts or accessible history reduce reliance on the removed record. Across 128 tasks per API configuration, removing memory content while preserving transaction identifiers reduces success by 0.656 [0.594, 0.715] and 0.566 [0.527, 0.605], respectively. Content replacement redirects actions toward the altered record, distinguishing information use from correctness. In software localization, a frozen repetition-intervention diagnostic evaluated on 800 new issues across four models attains an area under the receiver-operating-characteristic curve (AUC) of 0.9403 versus 0.9365 for a token-funded generic comparator. The paired difference is 0.00382 (95% interval [0.00023, 0.00743]); the prespecified family-adjusted interval includes zero. These findings connect controlled state responses to recoverable use of agent-written information and specify the conditions under which saved records contribute to execution.

Keywords:

language agents, agent evaluation, state-action binding, causal interventions, reliability, SWE-bench

1. Introduction

Language agents use memories, plans and persistent goals to carry information from one interaction to the next (Yao et al., 2023; Shinn et al., 2023; Park et al., 2023). A recorded

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budget can constrain a later purchase; a scheduling policy can determine which of several feasible appointments to select. In each case, the important relation is between what the agent retains and what it subsequently does. A successful transaction establishes that the task was completed. Explaining the contribution of a saved record requires comparing actions when that record changes while the execution context remains fixed.

We investigate this relation through *causal state binding* (CSB): the dependence of subsequent action on task-relevant exposed state, evaluated through targeted changes, irrelevant perturbation, restoration and alternative access to the same information. The central finding is that agent-written memories and goals make substantial, recoverable contributions to tool execution in the tested API configurations. Removing remembered task content still reduces success when account and object identifiers remain available. Replacing that content redirects actions toward the altered record, showing how the information retained by the agent shapes the transaction it executes.

Three distinctions organize the analysis. Information can be present without being used; an action can use a record that contains an incorrect value; and the same information can reach execution through memory or an alternative source. These distinctions connect memory-based action to research on the causal role of intermediate representations. Interventions on reasoning traces and neural representations have shown why their availability or plausibility alone is insufficient evidence of their use (Turpin et al., 2023; Lanham et al., 2023; Geiger et al., 2021). For an agent acting through tools, the saved record supplies an observable interface at which these possibilities can be separated experimentally.

The experiments connect controlled responses under explicit information rules to parameterized transactions after the agent has written its own state. A checkpoint between writing and execution holds the realized record and environment constant across intervention branches. Deletion, recovery, content separation and directional replacement form the central behavioral evidence (Figures 3–5). The software-localization analysis then asks whether repeated and intervened responses help predict success beyond generic diagnostic queries with comparable resources. This application links the behavioral study of state use to the practical assessment of an agent’s answers.

2. Related work

Task benchmarks evaluate agents against externally specified objectives. AgentBench, WebArena, τ -bench and SWE-bench cover interactive decisions, policy-governed tool use and software repair (Liu et al., 2024; Zhou et al., 2024; Yao et al., 2025; Jimenez et al., 2024). ToolSandbox evaluates intermediate and final milestones in environments with stateful tool dependencies (Lu et al., 2025). These settings make execution outcomes observable, including failures that plausible verbal answers can conceal. Our experiments use such outcomes to compare actions from a common checkpoint under different versions of a saved record.

Memory evaluation further distinguishes retaining information from applying it. Mem2ActBench constructs tasks whose tool parameters depend on facts distributed across prior interactions, directly testing memory-supported parameter grounding (Shen et al., 2026). Analyses that vary writing and retrieval strategies locate different bottlenecks in access to stored information and its utilization (Yuan et al., 2026). The present comparison fixes the record actually written by an agent and varies its availability, content and source of

access. This permits task facts to be separated from transaction identifiers and tests whether earlier history or redundant facts can substitute for the removed record.

Intervention studies of reasoning supply the closest methodological basis. Truncation, inserted mistakes and paraphrases test different forms of dependence on a chain of thought (Lanham et al., 2023); mediation analyses distinguish the effects of a reasoning problem and its intermediate reasoning on an answer (Paul et al., 2024). Interchange interventions test a more specific relation: whether replacing a representation produces the behavior associated with the substituted value (Geiger et al., 2021). We apply this directional logic to exposed memory and goals. Scoring each action against both the original and altered task target distinguishes movement toward the replacement from an undirected decline in success. The intervention concerns the record supplied to execution, with the continuation policy held fixed.

Controlled replay provides a complementary account of agent behavior. Causal Agent Replay estimates shifts in outcome distributions after interventions at selected steps and treats policy sampling as part of the comparison (Shah, 2026). Process-evaluation and executed-replay analyses likewise separate action prediction, task uncertainty and step contribution (He et al., 2026; Zhang, 2026). Our sampling unit is a task with a saved checkpoint: each condition receives fresh continuations from the same environment. Relevant deletion, irrelevant edits, restoration and content substitution characterize the distribution of executable outcomes conditional on the saved state and continuation policy.

Reliability estimation asks what generated responses reveal about correctness. Self-consistency aggregates repeated answers (Wang et al., 2023); semantic uncertainty groups answers by meaning before measuring their variation (Kuhn et al., 2023; Farquhar et al., 2024); SelfCheckGPT uses agreement across sampled generations as evidence about factuality (Manakul et al., 2023). These methods motivate a demanding comparator: extra generic observations can themselves be informative. The localization analysis therefore gives intervention and generic diagnostics matched information, learner-selection opportunities and recorded token funding. Development and independent evaluation are separated to distinguish representation selection from generalization (Cawley and Talbot, 2010). File localization and patch resolution remain distinct outcomes, following the separation of these tasks in software-repair systems (Xia et al., 2024).

3. Definition and study design

3.1. State dependence and intervention contrasts

For task e , let h_e denote the context supplied at the execution checkpoint, x_e the resettable environment, and $s_e = (m_e, g_e, u_e)$ the agent-written memory, persistent goal and unrelated record. Intervention I_b changes the designated state component before a fresh continuation with model randomness ϵ . Its scored outcome is

$$Y_{eb} = W_e f\{\text{Run}_\pi(h_e, I_b(s_e), x_e; \epsilon)\}, \quad (1)$$

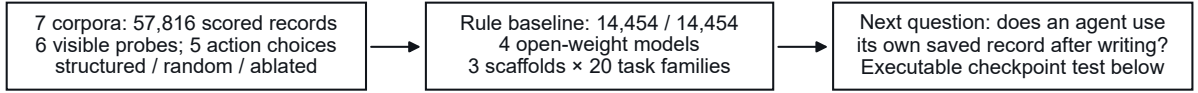
where f is an executable evaluator hidden from the model and W_e is the indicator of a valid state write. Raw environment success is retained separately from this preparation-gated primary endpoint. For component k , the primary paired effect is

$$\Delta_k = \mathbb{E}_e \left[\Pr(Y_{e,\text{sham}} = 1) - \Pr(Y_{e,\text{lesion}(k)} = 1) \right]. \quad (2)$$

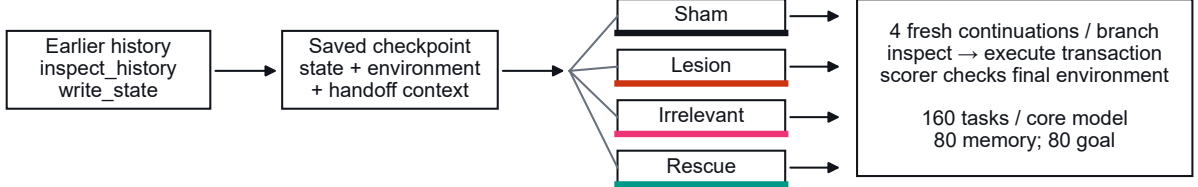
The expectation gives equal weight to the eight task mechanisms and then to their registered instances. The intervention holds the checkpoint, environment and continuation policy fixed. Restoration measures recovery after replacing the removed component with the agent’s original record; irrelevant deletion measures sensitivity to other context. Restated facts and accessible history test whether alternative information can compensate for the loss. These comparisons address different explanations for a deletion effect, including loss of task content, removal of required identifiers and broader disruption of the interface. Their interpretation also depends on state-write quality and baseline task competence. Four continuations per condition provide a discrete estimate of each task’s success probability.

Dependence is conditional on the information accessible at execution. When a fact is available only in saved memory, removal tests reliance on that source; when the same fact is restated at handoff, removal tests whether another source can compensate. A substituted value addresses a different question: whether the direction of action follows the changed content. Relevant dependence, specificity, recovery and alternative access therefore characterize complementary aspects of state use. Figure 1 connects these comparisons to the controlled-action and prediction studies. Finite-action response indices summarize choices under explicit rules; the action-field coupling index (AFCI) summarizes reported content with knowledge of its source; localization features quantify relations among predicted file sets. Their roles follow the outcome and information available to each evaluator. CSB names the state-action property that motivates these measurements.

a Controlled information: explicit rules and matched prompt controls



b Execution: intervene after the agent has written its state



8 mechanisms x 20 tasks; memory / goal = 80 / 80; repeats do not enlarge task n.

c One saved checkpoint, four observed continuations

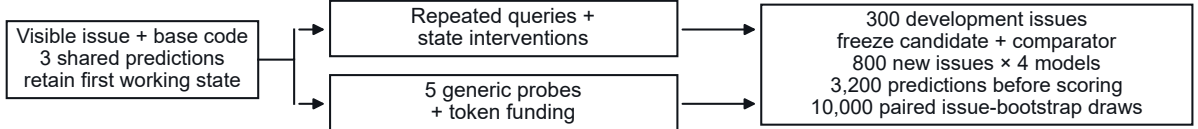
Sol / aggregation / memory 1: AC-407423; REP-9366

Memory: settled entries ≥ 100 cents. Goal: sum by day.

Stored memory	Mon / Tue / Wed (cents)	Original / altered target
Unchanged	750 / 900 / 800	1 / —
Empty dictionary	No accepted transaction	0 / —
Restored	750 / 900 / 800	1 / —
Threshold → 200	650 / 800 / 800	0 / 1

Same environment reset; independent draws. First nonredundant aggregation-memory checkpoint.

d Prediction: isolate reference-free features from outcome scoring



Reference paths: scorer only, released after all 3,200 predictions are frozen.

Figure 1: From controlled responses to state-dependent action. **a**, Explicit finite-action rules and matched controls establish the controlled response patterns. **b**, Agents inspect earlier history and write their own memory and goal before a checkpoint is saved. Each branch starts from the same environment with fresh model draws; 160 tasks per core configuration comprise 80 memory and 80 goal tasks. **c**, Recorded repeat-zero continuations from the first nonredundant aggregation-memory checkpoint in Sol (task index 1). The table reports unchanged, removed, restored and substituted memory under the same goal. The substituted record raises the amount threshold from 100 to 200 cents; original and altered targets are scored separately. Dashes mark the branches for which only the original-target score is displayed. This worked example illustrates the intervention semantics, while Figures 3–5 report all tasks. **d**, Predictors are selected on 300 development issues and frozen before 800 new issues are evaluated by four configurations; all 3,200 predictions precede reference-data release.

Table 1 distinguishes the state properties and the evidence used to evaluate them.

Table 1: State variables and measured outcomes.	
Term	Meaning in this study
Exposed state	The saved record supplied to the continuation at the execution checkpoint.
Memory	Task facts retained from the earlier interaction.
Persistent goal	A saved optimization or conflict-resolution policy that selects among feasible actions.
Execution success	Correct final local environment transaction, including parameters and task policy.
Reference-free diagnostic	Computable from visible inputs and model outputs, without reference answers.
AFCI	A score combining reported content, its source and final-action alignment.

3.2. Agent-written state and tool execution

The registered design contains eight mechanisms: service allocation, meeting scheduling, directed routing, inventory splitting, invoice arithmetic, dependency scheduling, versioned record merging and grouped event aggregation. The agent inspects earlier history and writes a durable state. After handoff it must inspect the operational environment and execute a parameterized tool transaction. The evaluator checks stock, intervals, paths, amounts, job dependencies or resulting records as appropriate. The synthetic tasks share a compact JSON tool interface, with diversity defined by these eight computational mechanisms.

Development used 20 initial allocation instances plus 20 instances spanning the eight mechanisms. Both development sets were separate from the 160 study instances, equally divided between memory and goal interventions. Each core branch used four fresh continuations. Thirty-two study instances restated the relevant information at handoff to test redundancy. A fixed 40-instance subset added semantic paraphrase, one-field replacement and accessible-history compensation. Sham and rescue contained identical saved state but used independently sampled continuations. Rescue restored the agent’s own written state, preserving its original values.

The API request identifiers are `gpt-5.6-sol` and the DeepSeek route `deepseek-v4-pro`; the fixed-weight core uses Qwen2.5-14B-Instruct and Mistral-7B-Instruct-v0.3 using the Q4_K_M weight-quantization format. The provider-served `gpt-6-astra` is allocated a fixed 40-instance subset, shared with the core models. API configurations are identified by their request routes and returned identifiers; open-weight configurations are identified by immutable repository revisions and file hashes. The execution models use a different quantization from the controlled open-weight experiments, so comparisons across those studies also differ in numerical precision.

Fixed-weight inference used the pinned llama.cpp implementation with 15,360 context tokens per slot and quantized Q8 key/value caches. Inference used the same weight files and scientific code across the recorded backends, with independently sampled continuations. Requests set temperature 0.7 and top- $p = 1$; the recorded fixed-weight defaults additionally

retain top- $k = 40$ and min- $p = 0.05$. Fixed-weight seeds derive from request identifiers; API continuations follow provider sampling. Reasoning effort was set to low for the GPT API configurations. The common JSON interface validates responses as returned. Preparation and software queries allow up to 1,536 output tokens; execution continuation steps allow 1,024. Full prompts, caps, returned usage and runtime defaults are archived.

3.3. Reference-free issue-localization prediction

The visible interface contains only issue text, repository identity, immutable base commit and task role. Retrieval ranks implementation paths from that commit and supplies fixed source excerpts. Reference patch paths, test outcomes and labels remain in a separate scoring file. The three shared base calls predict up to three files and write a working hypothesis; the first call’s state is retained without selecting it by confidence or correctness. The binary endpoint is at least two of the three base calls each retrieving at least one reference implementation file in its top three.

In the fixed linear comparison, each strategy receives the same visible issue/code information and the same original working state, with three shared calls plus five strategy-specific calls. The state probes are current state, minimal candidate-file state, removed state, an irrelevant note and a synthetic mismatch drawn from the same visible candidate pool. Generic candidates were five additional self-consistency calls and five instruction paraphrases. Each augmentation supplies seven features to an equally specified logistic predictor. The prespecified development rule selected self-consistency before formal collection: paraphrase improved its own grouped development AUC by 0.008, below the 0.020 threshold for switching. This selection did not compare either candidate against CSB.

Let $J(A, B) = |A \cap B| / |A \cup B|$, with empty or invalid file sets assigned similarity zero, and let C be the top-three file-vote consensus from the base calls. Writing $P_c, P_m, P_{\text{removed}}, P_i, P_x$ for the current, minimal, removed, irrelevant and mismatched-state output sets, the first five state features are $J(P_c, C)$, $J(P_c, P_m)$, $1 - J(P_c, P_{\text{removed}})$, $J(P_c, P_i)$ and $1 - J(P_c, P_x)$. The two difference features are zero if either output is invalid. The remaining two features are the invalid-output fraction and mean reported confidence. Generic features use mean agreement with C , pairwise agreement, normalized vote margin, path-vote entropy in nats, mean base-to-extra agreement, invalid fraction and mean confidence. Unavailable confidence values receive training-fold median imputation.

The development cohort contains 300 SWE-bench Lite test issues, also used by the six-model reference-dependent study and called the bridge in the figures. Twenty issues from the separate Lite development split were used to validate the implementation and select the generic comparator. A 100-issue SWE-bench Verified cohort excludes duplicate IDs, repository/commit matches and registered text/patch near-duplicates. Its previously observed outcomes support a reassessment of the amended model cohort, with labels excluded from fitting. The Astra comparison uses a fixed 60-issue development subset. Supplementary Methods describe the cohort composition and exclusion rules; the research package provides the issue-level membership lists.

The independent comparison used richer reference-free features and a hybrid (repetition–intervention) diagnostic: three current-state reevaluations, one removed-state response and one mismatched-state response. Generic strategies received the same visible information, representation choices and learner-search opportunity. After development on the 300 issues,

all predictors and analysis choices were fixed. We selected 800 unused issues from the remaining full SWE-bench test population, stratified across 12 repositories, after excluding previously used IDs and near-duplicates. The full test population differs from the Lite and Verified subsets used for development and reassessment. Predictions for all four model cohorts, comprising 3,200 records, were fixed and their hashes recorded before the scoring file was released. The 800-issue cohort thus provides a separate test of the frozen predictors.

4. Results

4.1. *Structured information supports controlled action responses*

Structured information produced a consistent action-response advantage in the finite-action experiment (Figure 2). Across 57,816 scored records from seven corpora, the structured condition exceeded the high-randomness comparator in every corpus under the registered composite criterion. The mean contrast was 0.886, with a descriptive two-sided 95% Student t interval of [0.843, 0.929] across the seven corpus differences. Reason and veto removal also changed their corresponding responses in the registered directions. The corpus is the inference unit for these comparisons; shared templates and scoring rules define their common experimental setting.

The separate four-model control matrix extends the comparison to Qwen2.5-7B, 14B and 32B and Mistral-7B. Structured inputs outperformed context-field removal in 238/240 scaffold-task-family cells, scrambling in 240/240 and the distribution-matched control in 236/240, with mean contrasts of 0.550, 0.895 and 0.593. Ordinary state-field deletion produced a smaller mean contrast of 0.041, positive in 193/240 cells. Strict decisive-field deletion in the additional Qwen-32B experiment produced a mean contrast of 0.961, positive in all 60 cells. These cell summaries show how the effect depends on which information the intervention removes; they are descriptive comparisons within the shared design.

The visible rules provide a direct account of performance in this setting. A deterministic baseline, given the same rule template and module flags, reproduced all 14,454 structured-condition targets, with accuracy 1.000. The scorer’s target variable was withheld from generation, but the correct response was recoverable from the visible rule. Thus, the finite-action experiment establishes dependence on explicitly supplied information. The tool-execution experiment addresses the corresponding question for records produced by the acting agent itself.

The five available responses were action A, action B, veto, deferral and recall of a prior commitment. Malformed or unmatched responses were recorded as invalid. Table 2 gives the probe-to-action mapping; deferral is permitted by the interface but is not a target of these six probes. The synthetic open-weight control matrix uses a separate four-action set. The canonicalization procedure and response-score definitions are given in Methods.

Table 2: Visible prompt-to-action rules in the finite-action experiment. The module flag and rule are available to both the model and the deterministic baseline.

Probe	Relevant module enabled	Module disabled
Baseline / irrelevant cue	<code>ACTION_A</code>	<code>ACTION_A</code>
Reason flip	<code>ACTION_B</code>	<code>ACTION_A</code>
Memory conflict	<code>RECALL_PRIOR</code>	<code>ACTION_A</code>
Veto cue	<code>VETO</code>	<code>ACTION_A</code>
Self-continuity	<code>ACTION_A</code>	<code>ACTION_B</code>

A decisive-field positive control further tested the information responsible for the choice. It comprised 1,440 API generations across 20 task families, six events per family, six conditions and two models. Both full-state and decisive-field-only accuracy were 1.000 for GPT-5.4-mini and Qwen3-max; the strongest non-decisive control, action-prior-only, reached 0.208. A local consistency guard checked the event identity of the decisive field and its agreement with the action, correcting locally or deferring when the field belonged to another event. Without additional model calls, following of scrambled fields fell from 0.958 to 0.033 in GPT-5.4-mini and from 0.533 to 0.033 in Qwen3-max, while irrelevant-cue accuracy remained 1.000. The guard demonstrates a concrete use of event-specific information where the action rule is explicitly available.

The corpus distributions, replicate values and leave-one-corpus summaries are reported in Supplementary Figures S4–S5; Figure S9 gives the complete open-weight comparison. The individual corpus and replicate values show the variation underlying the aggregate contrasts.

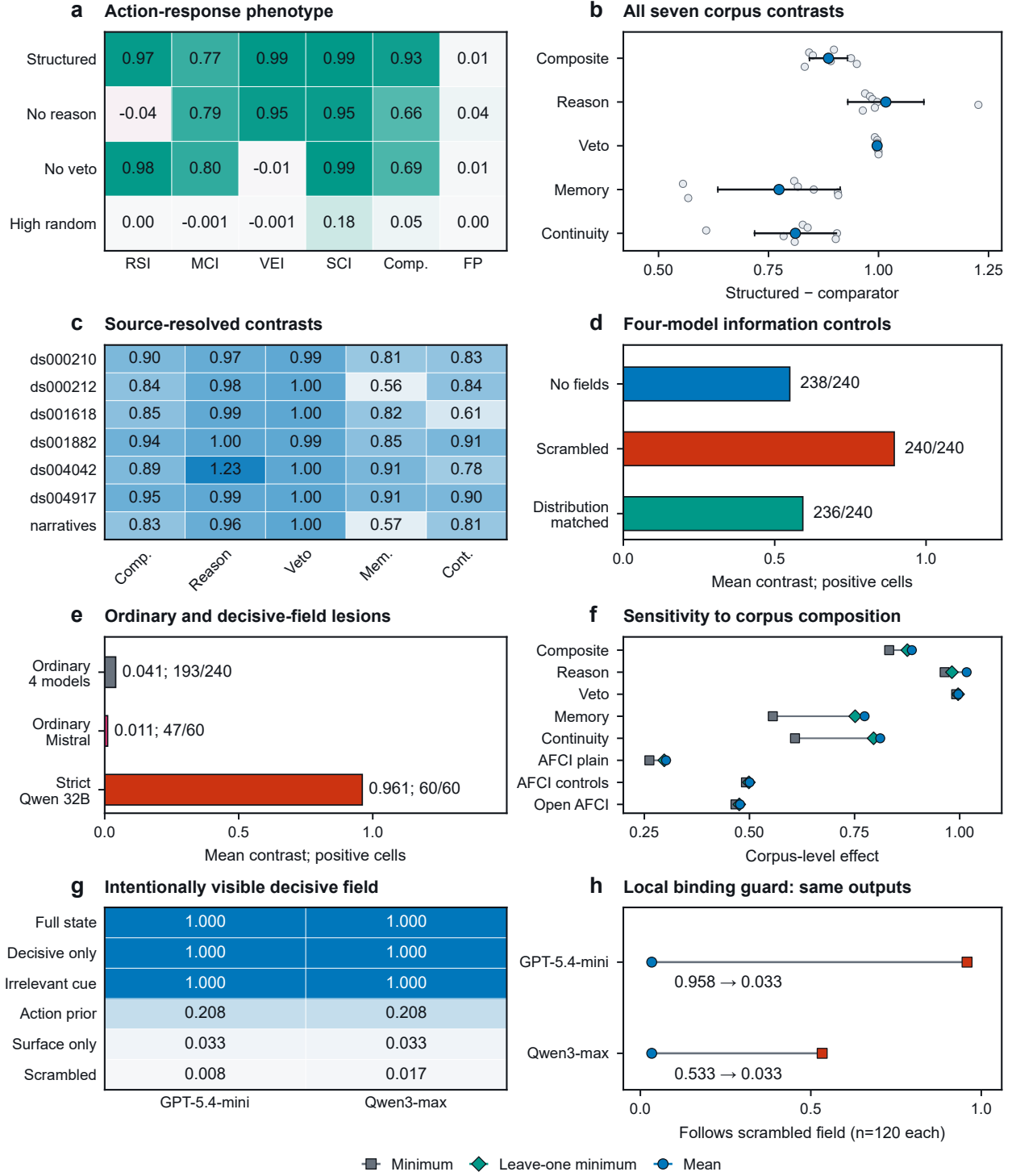


Figure 2. Controlled information and action responses.
Panel b: seven-corpus response-score contrasts with descriptive 95% Student t intervals ($df = 6$).

Figure 2. Detailed legend.

Controlled information and action responses. **a**, Behavior-level reason (RSI), memory (MCI), veto (VEI), self-continuity (SCI), composite and irrelevant-cue false-positive (FP) scores. **b,c**, The same five response-score contrasts across seven corpora, summarized in b and resolved by corpus in c. Reason and veto compare structured with the corresponding component deletion; composite, memory and continuity compare structured with high-random output. Differences have range $[-2, 2]$. Descriptive two-sided 95% Student t intervals (6 degrees of freedom) and source heterogeneity; the original percentile-bootstrap criterion is reported separately in SI Table S45. **d,e**, Open-weight mean contrasts and positive scaffold-task-family cells; strict deletion is a separate Qwen-32B slice. **f**, Minimum, leave-one-corpus minimum and mean effects; colors and shapes follow the bottom key. **g**, Decisive-field positive-control accuracy (120 events/condition/model). **h**, Scrambled-field following before (red square) and after (blue circle) the local guard, on the same 120 outputs/model without an added call. Cell counts are descriptive; corpora share templates. Full profiles remain in SI Figures S4, S5 and S9.

4.2. Agent-written memory and goals support recoverable execution

Removing saved memory or a persistent goal reduced transaction success in Sol and DeepSeek (Table 3; Figure 3). The four core configurations received the same 160 tasks, divided equally between the two components, with four fresh continuations per condition. Sol’s memory and goal effects were 0.750 and 0.366; DeepSeek’s were 0.541 and 0.259. All four passed the registered eight-comparison Holm adjustment ($p < 0.001$), and their unperturbed success rates ranged from 0.675 to 0.938.

The quantized open-weight configurations had lower baseline competence. Qwen’s effects were 0.081 for memory and 0.044 for goal (adjusted $p = 0.031$ and 0.137), with unperturbed success rates of 0.106 and 0.125. Mistral completed only 1/320 unperturbed memory continuations and 4/320 goal continuations. Its small contrasts therefore occur at floor performance, where changes in an already rare success event have limited diagnostic resolution.

Table 3: Execution after agent-written state. Effects are sham minus targeted-lesion success with equal mechanism weights. Intervals resample tasks within mechanisms; sham is the unweighted observed success rate. All planned preparation and execution failures are included.

Model	Component	Tasks	Sham	Effect	Two-sided 95% CI
Sol	memory	80	0.925	0.750	[0.656, 0.834]
Sol	goal	80	0.938	0.366	[0.269, 0.462]
DeepSeek route	memory	80	0.675	0.541	[0.453, 0.622]
DeepSeek route	goal	80	0.681	0.259	[0.184, 0.338]
Qwen 14B	memory	80	0.106	0.081	[0.034, 0.134]
Qwen 14B	goal	80	0.125	0.044	[0.009, 0.084]
Mistral 7B	memory	80	0.003	0.003	[0.000, 0.009]
Mistral 7B	goal	80	0.013	0.006	[0.000, 0.019]

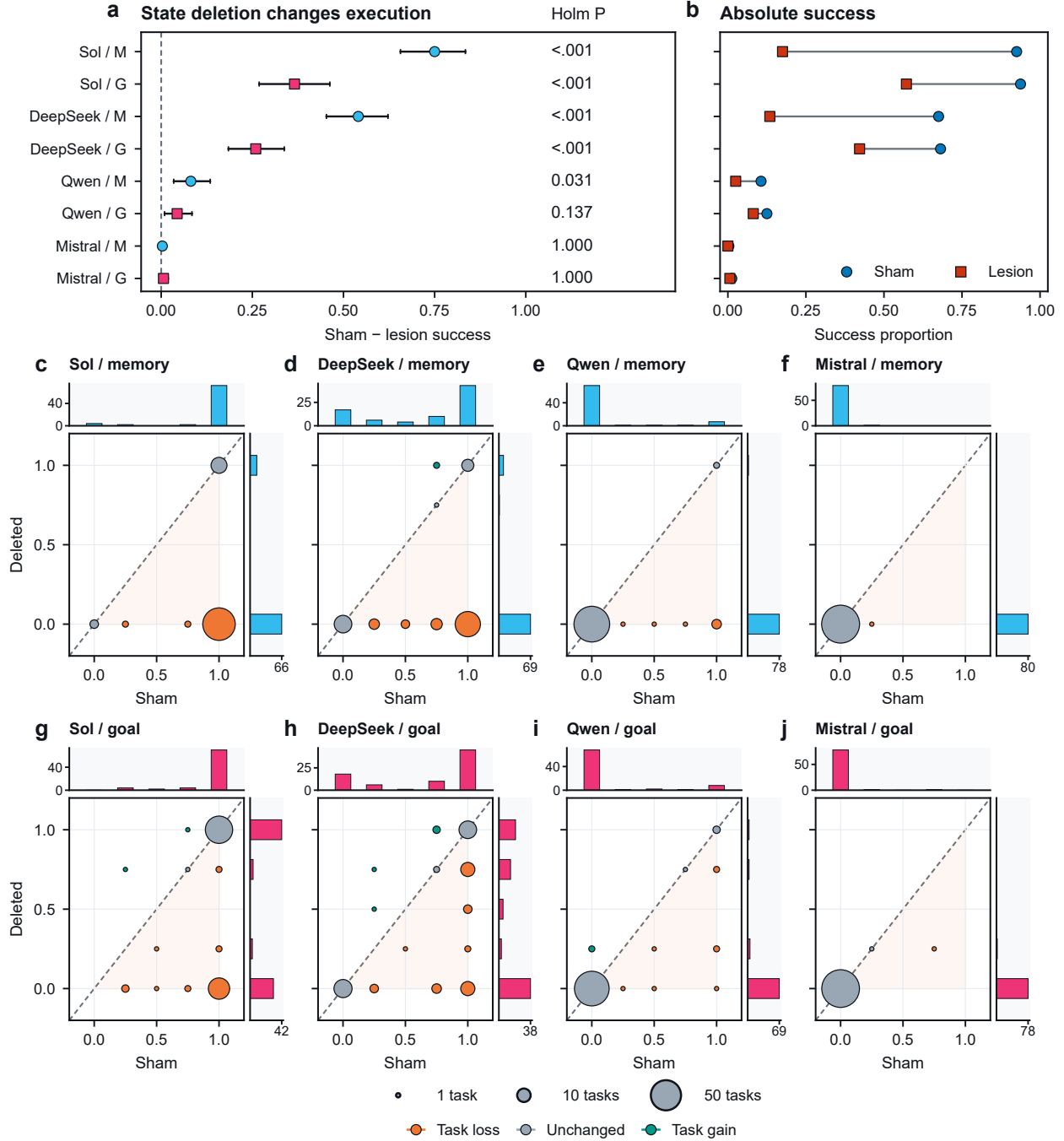


Figure 3: State dependence, absolute competence and paired task outcomes. M and G denote memory and persistent goal. **a**, Registered equal-mechanism sham-minus-lesion effects and two-sided 95% task-stratified intervals (80 tasks per component/model), with Holm adjustment across eight primary contrasts. **b**, Absolute sham and lesion success over 320 continuations each. **c–j**, All 80 paired task means per panel, averaging four continuations per branch; the first row is memory and the second persistent goal. Bubble area is proportional to task count at each attainable pair of success fractions. Orange marks lower lesion success, teal higher lesion success and gray ties; the dashed diagonal denotes equality. Aligned top and right histograms show the respective marginal task counts. Cyan and magenta identify memory and persistent goals in **a** and in the marginal histograms. The baseline success rates in **b** provide the context for the deletion effects. All planned cases, including failures, remain in the denominator. The task is the paired analysis unit. SI Figures S1–S3 resolve mechanisms, matched Astra subsets and failure intersections.

Restoring the agent’s own saved record recovered most of the success lost under deletion. Rescue-minus-lesion effects for memory and goal were 0.753/0.366 in Sol and 0.519/0.238 in DeepSeek. Their unperturbed-minus-irrelevant differences ranged from -0.013 to 0.006 . The direct paired specificity comparison, irrelevant perturbation minus targeted deletion, further isolates this contrast (Figure 4). Memory effects were 0.744 [0.656, 0.828] for Sol and 0.538 [0.456, 0.619] for DeepSeek; goal effects were 0.359 [0.269, 0.453] and 0.272 [0.191, 0.353]. These four comparisons passed the separate supplementary Holm family ($p < 0.001$). Qwen showed a smaller memory effect of 0.078 [0.031, 0.128] ($p = 0.015$); its goal comparison gave $p = 0.090$, and neither Mistral comparison established a difference. The analysis specification and separate comparison family are given in Methods.

Dependence also varied with alternative access to the information. Providing earlier history increased memory-lesion success by 0.563 in Sol and 0.438 in DeepSeek. When relevant facts were already restated at the execution checkpoint, the corresponding memory-lesion effects were 0.000 and -0.016 . Together with restoration, these controls show that access to task-relevant information explains more of the observed pattern than the mere presence of a particular storage field. Paired descriptive intervals summarize the control residuals; encoding equivalence is evaluated separately in the content experiment.

Replacing saved content redirected transactions toward the corresponding alternative. We scored each original-state and replacement-state action against both the original and alternative task target. For memory, Sol actions matched the original target in 75/80 original-state continuations and 19/80 replacement continuations, while matches to the alternative target rose from 4/80 to 61/80. DeepSeek showed the same directional pattern: original-target matches fell from 57/80 to 7/80, and alternative-target matches rose from 3/80 to 56/80. The symmetric target-alignment contrasts were 0.706 and 0.644 for memory and 0.719 and 0.638 for persistent goals, respectively. These crossed comparisons identify the direction of state-dependent action alongside the deletion and restoration effects. Qwen showed little target-specific redirection and Mistral remained at floor performance. The supplementary analysis retains overlapping and undefined alternative targets and distinguishes redundant from nonredundant access to the original information.

The matched 40-task Astra comparison, mechanism-specific effects, preparation checks and failure intersections appear in Supplementary Figures S1–S3 and Notes 7–9. Figure 3 shows every task pair alongside baseline performance; Figure 4 compares recovery, specificity and alternative information access.

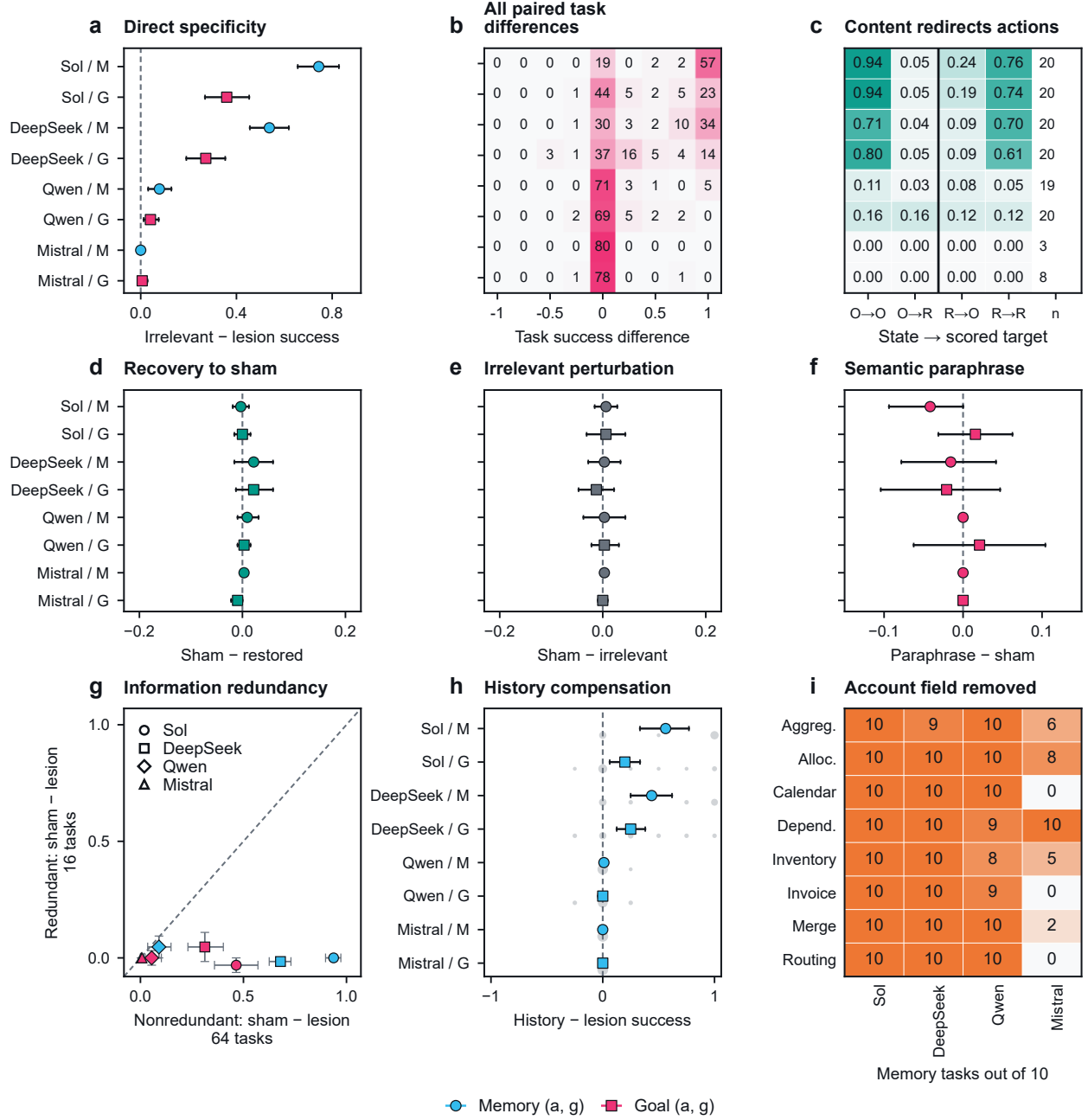


Figure 4: Specificity, recovery and the direction of state-dependent action. Memory (M) and goal (G) row order is shared in a–f and h. **a**, Irrelevant-minus-lesion effects with 95% intervals from 5,000 task-stratified supplementary bootstrap draws (80 tasks/cell). **b**, Exact task counts at the nine attainable paired differences; every row contains 80 tasks. **c**, Actions generated under original (O) or replaced (R) state, scored against each target: O→O, O→R, R→O and R→R correspond to $S_{00}, S_{01}, S_{10}, S_{11}$. The column n gives tasks with a defined alternative target, out of 20 planned; each contributes four continuations per state. **d,e**, Sham-minus-restored and sham-minus-irrelevant residuals (80 tasks). **f,h**, Paraphrase-minus-sham and history-minus-lesion contrasts on 20-task subsets; pale point areas in h show frequencies of individual task differences. Circles denote memory and squares persistent goals in these contrast panels. **g**, Nonredundant versus redundant effects with separate 95% intervals (64 and 16 tasks); colors distinguish components, shapes configurations, and the diagonal indicates equality. **i**, Memory records whose account field is removed by the original lesion, out of ten tasks per mechanism/configuration (266/320 overall). All displayed intervals are two-sided 95%. Figure 5 isolates memory content while preserving identifiers; its panel g compares encoding controls.

4.3. Memory content contributes beyond transaction identifiers

Saved memory supported execution through its task content as well as the identifiers required by the tool interface. We evaluated Sol and DeepSeek on 128 tasks from the eight mechanisms under eight memory conditions; task identities and analysis were fixed before acquisition. Each task had two continuations per condition. Retaining account, object and other identity fields while removing the remaining memory content reduced Sol success from 206/256 to 38/256 and DeepSeek success from 169/256 to 24/256. The paired probability differences were 0.656 (95% task-bootstrap interval [0.594, 0.715]) and 0.566 [0.527, 0.605], respectively; both two-model Holm-adjusted sign-flip tests gave $P < 0.001$ (Figure 5). Thus, the memory-removal effect persisted when transaction identifiers remained available.

The content-loss contrast was positive in all eight mechanism-specific point estimates, with magnitudes varying across mechanisms and models. Across the 128 paired tasks, identity-only memory reduced the observed two-continuation success fraction on 103 Sol tasks and 86 DeepSeek tasks; 22 and 40 were unchanged, and 3 and 2 increased, respectively. Identity fields also mattered: neither content-only nor empty-memory continuations produced a successful transaction in either model. Restoring the agent’s original record gave 215/256 successes in Sol and 167/256 in DeepSeek; irrelevant editing gave 215/256 and 166/256. Their paired differences from full memory were small relative to the content-removal effects. All 4,096 planned continuations were retained, including the consequences of one unsuccessful DeepSeek state write. The content-loss contrast remained positive under the extreme assignments of terminal service failures examined in the supplementary sensitivity analysis. Supplementary Methods report the complete failure accounting and these bounds.

Encoding controls separated the availability of facts from their presentation. Exact JSON text preserved the complete saved memory while changing its container from a structured object to a string. In Sol, it increased success by 0.074 [0.020, 0.129] relative to the structured record. Deterministic rewording preserved the same values; its difference from exact JSON text was -0.094 [-0.148 , -0.039]. In DeepSeek, the reworded-minus-JSN contrast was 0.008, with a 90% interval of [-0.027 , 0.042] inside the prespecified descriptive equivalence range of ± 0.05 . Rewording differed little from the original structured record in the overall point estimates (-0.020 in Sol and -0.012 in DeepSeek). The experiments therefore distinguish a large dependence on remembered task content from smaller, model-dependent effects of its encoding.

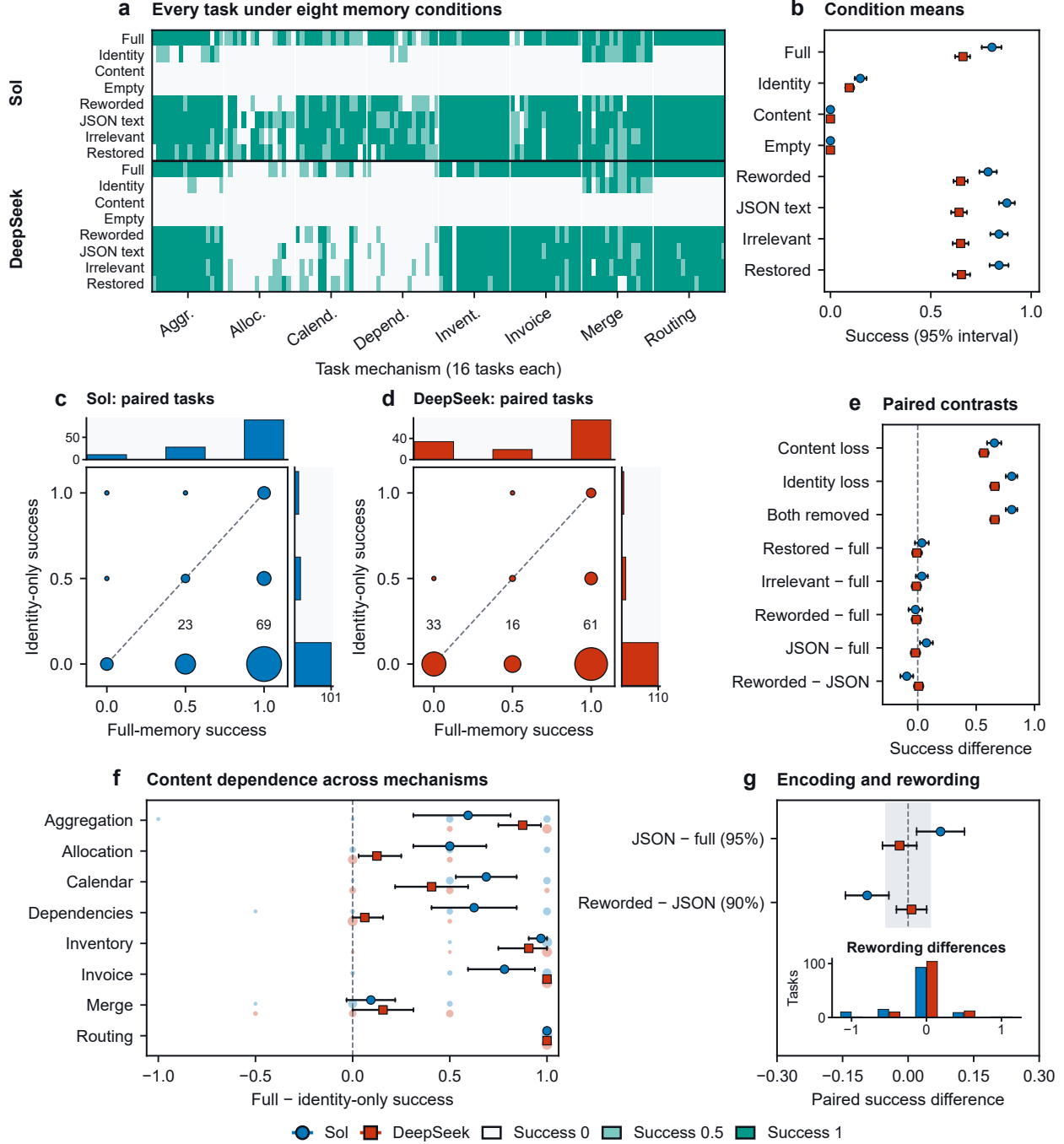


Figure 5: Memory content supports execution when identifiers remain available. **a**, All eight conditions for the same 128 new tasks, ordered within eight mechanisms. Separate Sol and DeepSeek blocks show two-continuation task means; colors encode success fractions 0, 0.5 and 1. **b**, Condition means with 95% task-bootstrap intervals (256 continuations/condition/model). **c,d**, Full-memory versus identity-only task means. Bubble area denotes task count on a common scale; cells with at least 16 tasks are labeled. Dashed lines indicate equality; marginal histograms retain all 128 tasks. **e**, Paired contrasts with 95% intervals. Intervals in **b** and **e** use 10,000 mechanism-stratified draws; the two primary content-loss tests use 50,000 sign flips with Holm adjustment across models. **f**, Mechanism-specific content-loss effects with descriptive 95% intervals over 16 tasks. Pale point areas show frequencies of attainable task differences. **g**, JSON-text-minus-full contrasts use 95% intervals; reworded-minus-JSON contrasts use 90% intervals for the registered descriptive equivalence comparison. The gray band marks ± 0.05 ; the inset shows all discrete task-level rewording differences. Denominators include failed preparations and terminal service failures; zero-width intervals describe floor observations.

4.4. State diagnostics and localization reliability

To evaluate reference-free prediction, we applied frozen diagnostics to 800 previously unused issues from 12 repositories, each evaluated by Sol, DeepSeek, Qwen and Mistral. The evaluation used the same binary localization endpoint and included all 3,200 planned records. The repetition–intervention diagnostic achieved AUC 0.9403 against 0.9365 for the token-funded generic comparator (Table 4). The paired difference was 0.00382, with a marginal 95% issue-bootstrap interval [0.00023, 0.00743]. Its prespecified family-adjusted 98.333% interval was $[-0.00055, 0.00821]$; the three-comparison superiority criterion was therefore not met. The independent estimate indicates a small pooled difference; its variation across configurations is examined below.

Feature comparisons on the 300-issue development cohort describe the information used by the predictors. Here, equally specified linear predictors gave AUC 0.9235 for state probes and 0.9245 for token-funded generic probes. Their paired full-refit difference was -0.00097 $[-0.01317, 0.00473]$. Richer response features and an equally specified four-family ensemble increased development AUC to 0.9480 for a combination of repeated queries and state interventions, compared with 0.9433 for the enhanced generic comparator. Each arm received the same feature-representation choices, learner families and tuning opportunities. For the hybrid diagnostic, the matched development ablation gave AUC 0.9433 with output-quality features and 0.9487 with quality plus answer relations; adding state correspondence alone did not recover that gain. The full ablations and their conditional uncertainty are reported in Supplementary Methods. Predictors selected from these development data were frozen before the independent evaluation. Supplementary Figures S12–S14 report the linear comparison and the 100-issue reassessment.

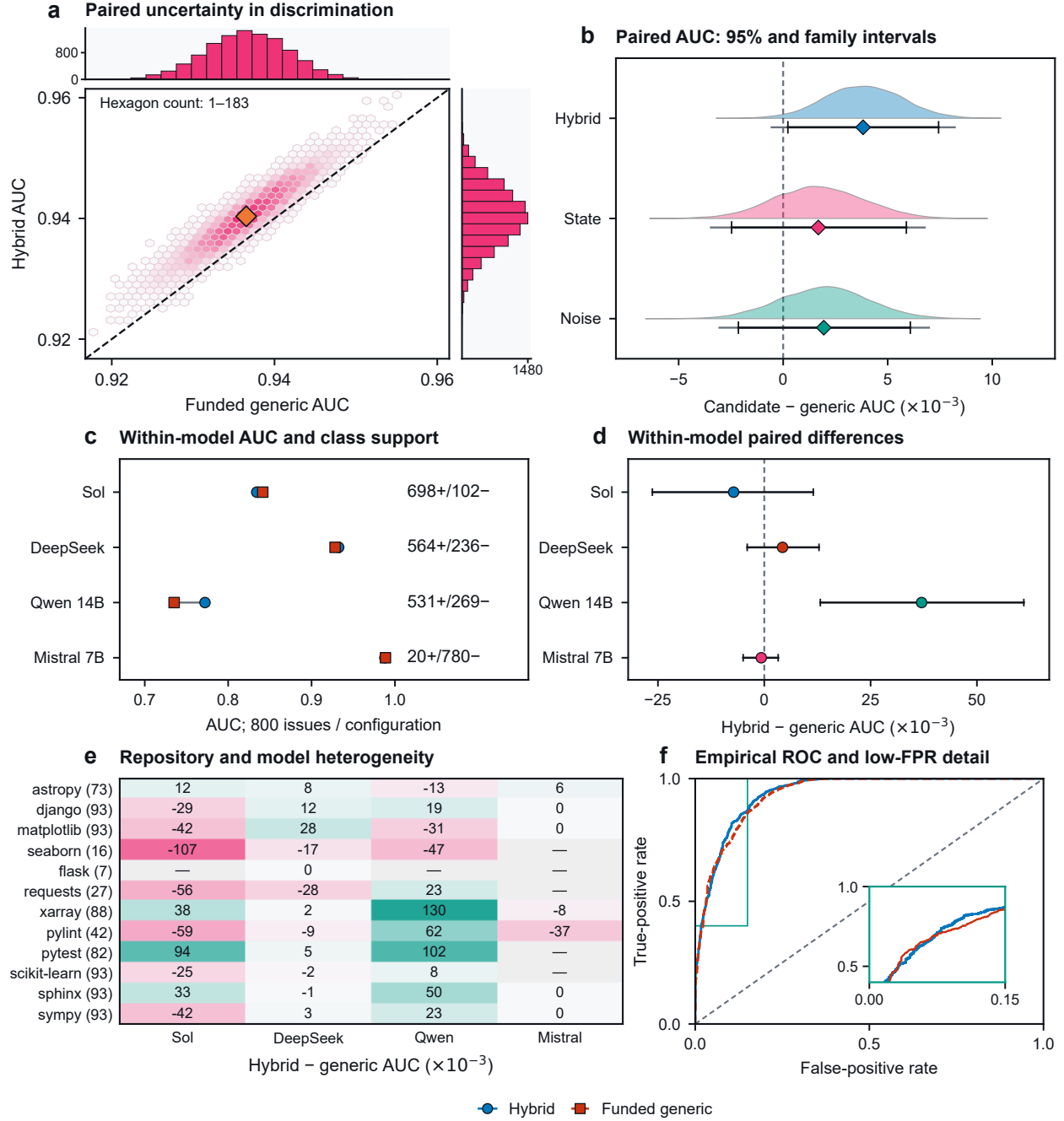


Figure 6: Independent prediction: paired discrimination and its heterogeneity. **a**, Joint distribution of hybrid and funded-generic AUC across the same 10,000 saved repository-stratified issue-bootstrap draws. Hexagon intensity and aligned histograms encode draw counts; the orange diamond marks the observed pair and the dashed diagonal equality. Each re-sampled issue carries its four model records together. **b**, Three prespecified comparisons with their funded generic predictors: repetition-intervention (primary), pure state and noise-contrast. Thin intervals give family-adjusted 98.333% coverage, thick intervals marginal 95%; normalized half-densities use the same saved draws. **c,d**, Within-model AUC pairs, positive/negative counts (800 issues/model) and primary differences with marginal 95% intervals. **e**, Descriptive hybrid-minus-generic AUC differences within each repository and configuration, in units of 10^{-3} . Teal and magenta indicate positive and negative differences on a common symmetric scale; row labels give issue counts. A dash marks a subgroup with a single outcome class, for which AUC is undefined. **f**, Pooled receiver-operating-characteristic curves with low-false-positive magnification. All predictors remain frozen. The primary family-adjusted interval crosses zero.

Table 4: Independent localization prediction on 800 new issues and four configurations (3,200 model–issue records). All predictors were frozen before the new outcomes were scored. Higher AUC and lower Brier score or log loss indicate better prediction. The primary paired AUC difference is 0.00382 (95% interval [0.00023, 0.00743]; prespecified family 98.333% interval [−0.00055, 0.00821]). Paired uncertainty and the comparison family are shown in Figure 6.

Diagnostic	AUC	Brier	Log loss
Five generic queries	0.9364	0.0927	0.2929
Token-funded generic	0.9365	0.0929	0.2936
State interventions	0.9382	0.0911	0.2901
Repetition–intervention	0.9403	0.0900	0.2873

Probability error showed the same overall direction. Brier score was 0.09000 for repetition–intervention and 0.09293 for funded generic diagnostics, a paired difference of −0.00293 with a marginal 95% interval [−0.00566, −0.00020]. The log-loss difference was −0.00624 [−0.01324, 0.00076]. These prespecified secondary measures characterize probability error alongside AUC. Figure 7 shows the fixed-bin calibration, actual probability support, paired losses and selective failure rates.

Model-specific effects distinguish the pooled result from a uniform gain. The primary AUC difference was 0.03699 [0.01318, 0.06103] for Qwen, 0.00430 [−0.00401, 0.01287] for DeepSeek, −0.00721 [−0.02631, 0.01151] for Sol and −0.00071 [−0.00496, 0.00327] for Mistral. These marginal subgroup intervals describe heterogeneity rather than an additional superiority family. Sol contributed 698 positive outcomes, DeepSeek 564, Qwen 531 and Mistral 20, each out of 800. The pooled AUC combines within-model and between-model differences; Mistral’s model-specific estimate is sensitive to its small number of successes.

The pure state-intervention ensemble and the noise-contrast variant gave AUC differences of 0.00168 and 0.00194 against their corresponding funded comparators. Both marginal and family-adjusted intervals crossed zero. The largest pooled point increment belongs to the specified combination of repeated observations and interventions; neither secondary comparison meets the superiority criterion.

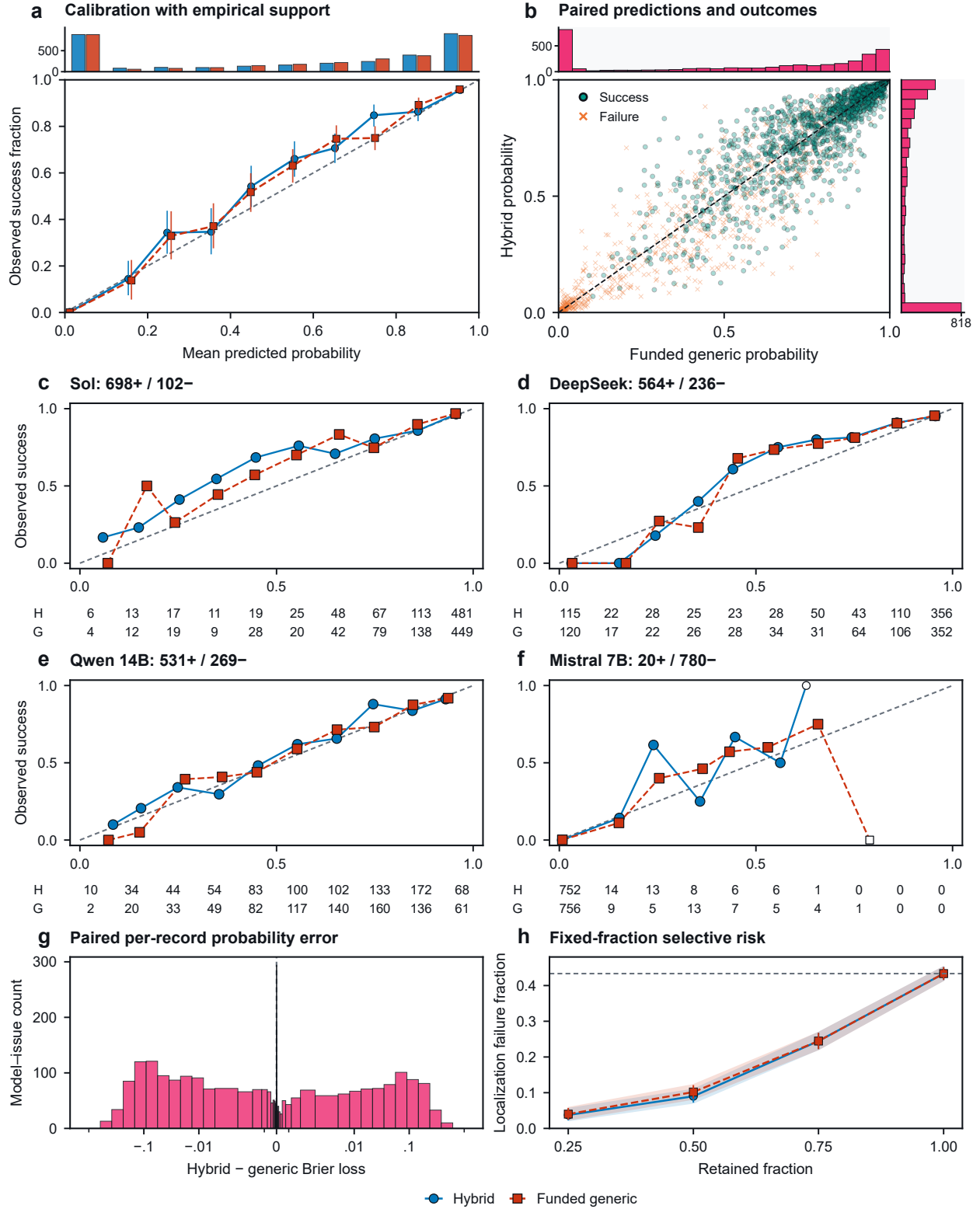


Figure 7. Calibration, paired probabilities and selective error on the independent cohort. 3,200 records; ten fixed bins; pooled and risk intervals: 95% paired issue bootstrap.

Figure 7. Detailed legend.

Calibration, paired probabilities and selective error on the independent cohort. **a**, Observed success versus mean predicted probability in ten fixed bins, with issue-bootstrap 95% intervals and an ideal-calibration diagonal. The aligned histograms give the actual bin counts. **b**, All 3,200 paired predictions, marked by teal circles for success and orange crosses for failure, with empirical marginal probability distributions; the diagonal denotes identical predictions. **c–f**, Model-resolved calibration; rows H (hybrid) and G (funded generic) give all ten fixed-bin counts, including zeros. Empty bins have no point; hollow markers identify bins with one record. These subgroup points are descriptive. **g**, All 3,200 paired per-record Brier-loss differences on a symmetric-log scale with linear threshold 0.005 and nonuniform bins; negative values favor repetition–intervention. **h**, Prespecified retained fractions of 25, 50, 75 and 100%, with failure rates and 95% issue-bootstrap intervals. Ties use the frozen deterministic rule, and risk intervals condition on the observed score selection. The new outcomes are used for evaluation throughout. Blue circles denote repetition–intervention and red squares funded generic diagnostics in a, c–f and h; panel b uses a separate outcome key.

Acquisition for the independent cohort recorded 241,102,491 tokens across all diagnostic strategies. Usage was complete and the generic comparator reached both candidate budgets in 3,121/3,200 records; 81 attempts lacked usage. In the complete-and-funded subset, the primary AUC difference was 0.00403 [0.00041, 0.00766]. Against the five-call generic comparator, the full-cohort difference was 0.00392 [0.00044, 0.00739]. Both additional comparisons showed a positive pooled AUC difference. Figure 8 displays all token pairs and the excluded accounting categories. The full planned cohort remains primary. Resource comparisons use reported tokens and distinguish complete from incomplete metering.

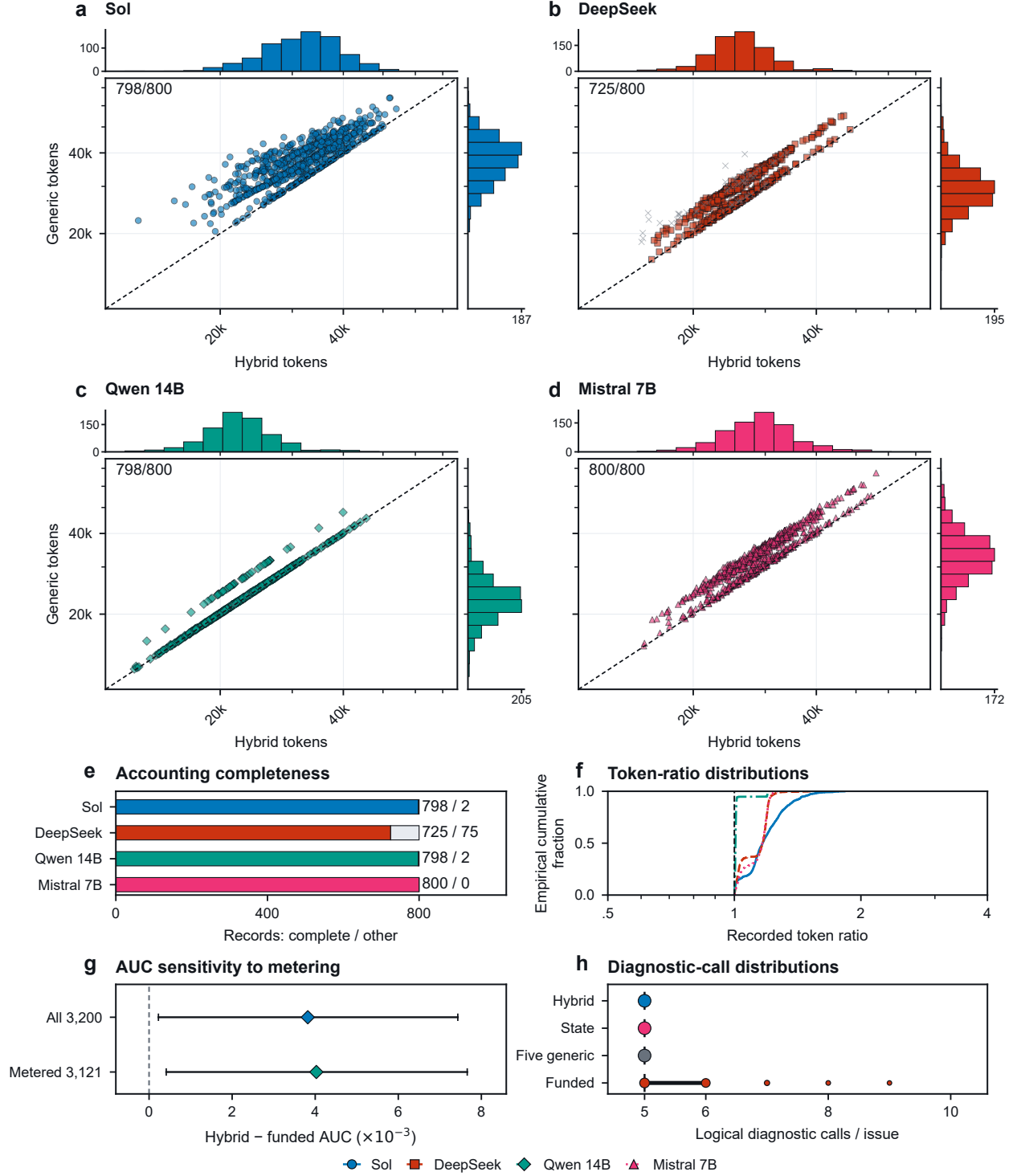


Figure 8: Resource pairing for the independent evaluation. **a–d**, Diagnostic tokens for all 800 model–issue pairs/configuration: repetition–intervention versus funded generic, with common logarithmic axes, equality diagonals and marginal histograms. Gray crosses mark incomplete usage; colored points are complete and funded. **e**, Corresponding counts by configuration. **f**, Empirical generic/candidate token-ratio distributions for positive recorded totals. **g**, AUC differences for all planned records (primary) and fully metered/funded records, with marginal 95% intervals. **h**, Logical diagnostic calls/issue: medians, interquartile ranges and attainable-count frequencies. Shared base calls are charged separately; costs include retries and token overshoot. Eighty-one attempts lack usage, affecting 79 records.

With reference implementation information included in the augmented features, a six-model analysis of 1,800 model–issue records gives AUC 0.737 for the ordinary baseline and 0.924 after augmentation, a difference of 0.187 [0.127, 0.268] under grouped validation. This comparison describes the association under reference-informed scoring; its feature information differs from that available to the reference-free predictors.

4.5. *Output agreement and the information available to a scorer*

The trace analyses distinguish alignment of reported content from its causal contribution to execution. AFCI averages reason, memory, veto and self-state alignment, using both the emitted fields and their generation provenance. Component values differ markedly: the structured self-state term is approximately 0.003, while the veto term is near 1. Figure 9 reports these components and 35 fixed sensitivity contrasts, comprising equal weights and each leave-one-component-out mean against seven comparators. Equal weighting defines an operational composite; the component profiles retain the information obscured by its mean.

Scoring without provenance reveals a separate constraint: the output must contain enough information for a judgment. A rule-based output-only analysis can score at least one component in 1.91% of 26,946 traces. An exploratory semantic assessment samples 252 outputs across nine variants and seven corpora and obtains 504 judgments from two models. Each judgment uses only emitted content and requires a verifiable quotation. Reason consistency is scorable in 79.4–86.5% of items, memory in 28.2–29.8%, persistent goal in 21.8–28.2%, and veto in 4.4–7.5%. Conditional consistency is generally high within those scorable subsets.

The judges also differ in whether they can score an item. With unscorable, inconsistent and consistent treated as three categories, inter-judge κ is 0.410 for reason, 0.810 for memory, 0.759 for persistent goal and 0.403 for veto. Only seven of the 252 outputs are scorable for veto by both judges. Reporting coverage alongside conditional consistency therefore preserves the distinction between observable agreement and the amount of evidence available.

The separate human evaluation gives high agreement on the seven visible-output dimensions. All 25,200 individual ratings were binary. Of 8,400 item–dimension combinations, 8,250 were unanimous and 150 had a 2:1 split, involving 68 items. Across the 21 annotator-pair-by-dimension comparisons, mean Cohen’s κ was 0.961, the minimum individual value was 0.860 and the minimum dimension mean was 0.897. Majority labels use at least two matching votes; agreement uses individual scores before aggregation. The two-annotator, 300-item evaluation uses a distinct sample. Supplementary Figure S10 reports prevalence and disagreement, and Figure S6 gives the complete AFCI profiles. These analyses characterize visible reports; the checkpoint interventions supply the evidence about effects on action.

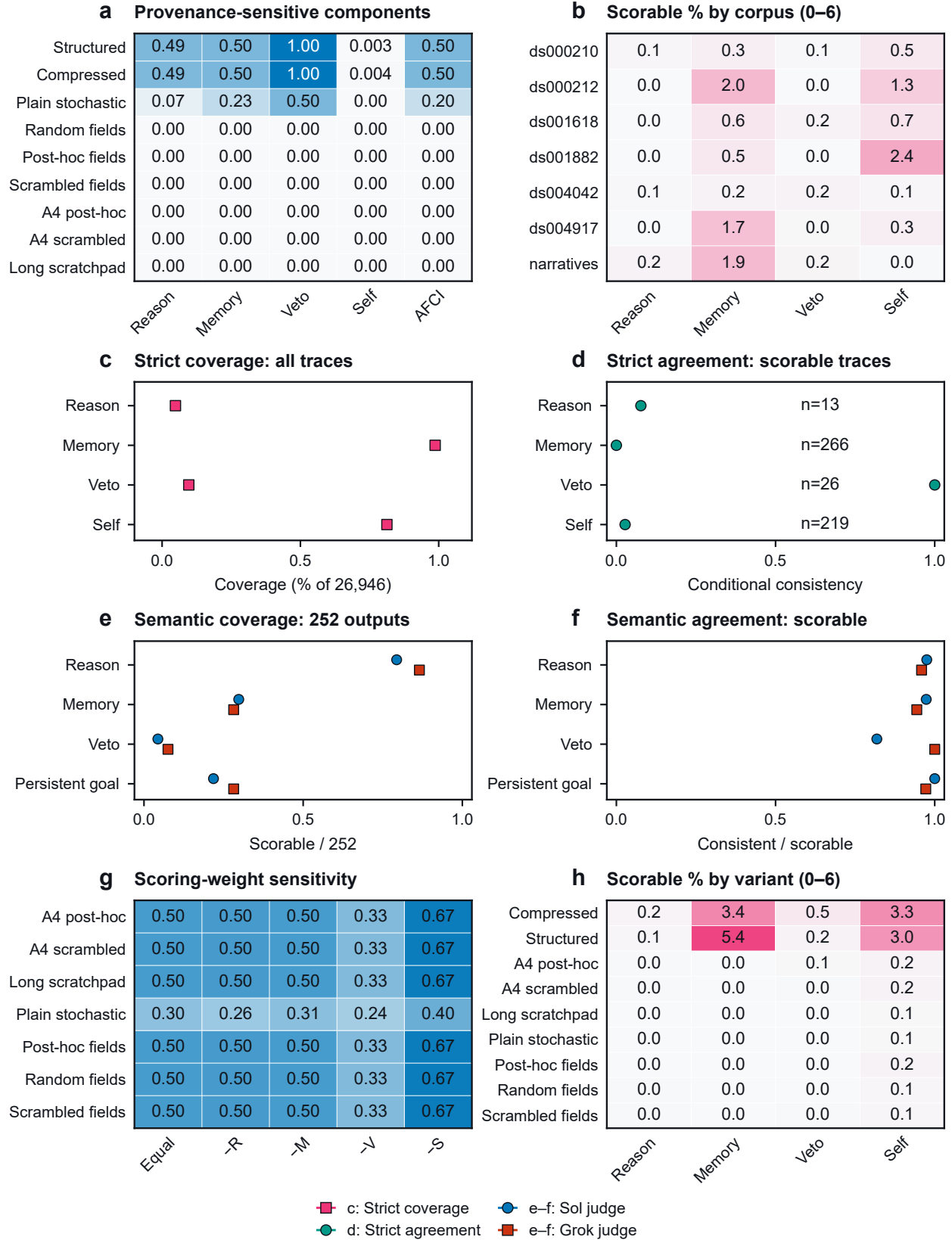


Figure 9. Measurement provenance, coverage and conditional consistency. 26,946 traces; two semantic judges on 252 outputs; descriptive scoring-weight contrasts.

Figure 9. Detailed legend.

Measurement provenance, coverage and conditional consistency. **a**, Content-and-provenance component scores and APCI, retaining three decimals for sparse self-state scores. **b,h**, Strict output-only scorable percentages by corpus and variant, from all 26,946 traces, with a shared 0–6% color scale. **c,d**, Strict coverage over all traces and consistency among scorable traces, on separate axes with the scorable counts printed in d. **e,f**, Coverage and conditional consistency from the Sol and Grok judges on 252 sampled outputs; the judge field named self evaluates explicit persistent-goal consistency. The denominators differ and these axes are not joined by lines. Legends distinguish strict scoring in c–d from semantic-judge identities in e–f. **g**, Equal-weight and leave-one-component-out APCI contrasts against seven controls, averaged over corpora; R, M, V and S denote reason, memory, veto and self-state. Weights were not optimized. Full replicate profiles and compression pairs remain in SI Figure S6. APCI and output-only consistency concern reported fields and their stated provenance.

4.6. Localization and patch resolution

The exploratory software-repair evaluation measures the distance between locating a relevant file and resolving the issue. A relevant file identifies where an edit may be needed; successful repair additionally requires an applicable patch and behavior that passes the repository tests. Of 100 initial model–issue submissions, 44 completed the official harness, one resolved the issue and 56 returned harness errors (Table 5). The local application-and-repair experiment began with 100 nonempty patches and made 93 repair attempts. Eight final patches, spanning seven unique issues, applied locally and completed the harness; none resolved its issue.

Table 5: Official repair evaluation. Counts are model–issue submissions; the filtered applicable subset is reported alongside the full initial denominator.

Stage	Submitted	Completed	Resolved	Harness errors
Initial: GPT-4o-mini	50	27	1	23
Initial: Qwen-plus-latest	50	17	0	33
Initial combined	100	44	1	56
After local apply screen/repair	8	8	0	0

The 92 zero-length final exports include rejected patches. Supplementary Note 4 reports format errors, context or hunk mismatches, missing paths, empty repair responses and other application failures by category. Localization and patch resolution are evaluated as distinct endpoints. A separate software binding-guard condition reduced hard-constraint violations by 0.166 while reducing primary hit@3 by 0.077, documenting a tradeoff between constraint enforcement and retrieval success.

4.7. Entropy and generation-budget controls

The local forced-budget experiment retained a structured-response advantage under exact prompt padding and fixed generation length: Qwen2.5-1.5B accuracy was 0.900 versus 0.325. This comparison fixes both the prompt and generation budgets. Across the larger entropy calibration, the smallest maximum gap was 1.171 bits, exceeding the registered 0.15-bit threshold, and prompt-token alignment did not establish equal total tokens and latency. Supplementary Figures S7–S8 report the full profiles and calibration attempts. These analyses distinguish the behavioral effects, exact local generation-budget control and entropy-calibration results.

5. Methods

5.1. Finite-action scores and action canonicalization

The finite-action evaluator maps raw output o_i to $\hat{y}_i = g(o_i)$ using the action set and visible rules in Table 2. It uppercases and trims text, normalizes the specified punctuation and hyphen variants, and applies ordered matches to the five response codes: `ACTION_A`, `ACTION_B`, `VETO`, `DEFER` and `RECALL_PRIOR`, including their aliases. The first match determines the action; unmatched output receives the evaluator code `INVALID_OR_UNMAPPED`. The scorer-side target is assigned before generation. It is fixed by probe condition across module variants and denotes the response expected when the relevant component is enabled. For example, the reason-flip target remains `ACTION_B` when a disabled-reason prompt instructs `ACTION_A`. The component score therefore measures preservation of the full-component response under intervention; compliance with each variant-specific instruction is a separate quantity. The deterministic baseline in Table 2 was evaluated on the 14,454 structured-condition records.

Let y_i^- and y_i^+ denote the expected actions before and after the manipulation, and z_i the probe condition. Correct response and irrelevant-cue change are

$$c_i = \mathbf{1}\{\hat{y}_i = y_i^+\}, \quad u_i = \mathbf{1}\{z_i = \text{irrelevant}\} \mathbf{1}\{\hat{y}_i \neq y_i^-\}. \quad (3)$$

For component k and condition v , the operational response score is

$$B_k(v) = \widehat{\Pr}(c_i = 1 \mid z_i \in \mathcal{T}_k, v) - \widehat{\Pr}(u_i = 1 \mid z_i \in \mathcal{N}_k, v), \quad (4)$$

where $\mathcal{T}_k$ and $\mathcal{N}_k$ are the component’s target and irrelevant/placebo probe sets. Each component score lies in $[-1, 1]$, so a between-condition score difference can lie in $[-2, 2]$. Reason, memory, veto and self-continuity scores use their corresponding probe definitions; their composite assigns equal arithmetic weight. The separate trace-based composite and AFCI definitions are given in Supplementary Section S8. The finite-action summaries in Results and Figures 2b and S4b use descriptive Student t intervals over the seven corpus differences: $\bar{d} \pm t_{0.975,6} s_d / \sqrt{7}$, where s_d is their sample standard deviation. The original registered criterion used corpus-cluster percentile bootstrap resampling, retaining within-corpus observations together; its saved 2.5th-percentile endpoint, 0.855732, is reported separately in Supplementary Table S45. The t interval describes the mean corpus contrast, whereas the percentile-bootstrap endpoint evaluates the registered criterion.

5.2. Execution effects and uncertainty

The task instance is the pairing unit. Preparation requires history inspection followed by a state write with three dictionary fields: memory, goal and unrelated material. Removing a component replaces its value by an empty dictionary while retaining the key. Continuation permits environment inspection followed by one transaction; branch and repeat order are deterministically shuffled within each checkpoint. Supplementary Methods provide the exact checkpoint structure and access rules. Each branch starts from the same saved environment and fixed handoff; different repeats use new model draws. The registered primary endpoint requires a valid state write and a successful final transaction. An invalid state write, malformed tool request, rejected transaction or missing response remains in the

planned denominator and yields primary success zero. Raw environment success is reported alongside the preparation-gated endpoint, including correct transactions following an invalid state write. State-write validity and component accuracy are reported separately. Analyses conditional on a valid write are secondary and explicitly selected. Unrelated-content deletion uses the agent’s existing note to match the removed serialized character count where possible. Serialized-character counts and tokenizer-based lengths are reported separately.

The fixed-suite primary analysis averages within-task differences with equal weighting of the eight mechanisms and reports memory and goal separately. Stratified task bootstrap intervals describe this suite. The primary analysis uses 5,000 stratified task-bootstrap draws and two-sided sign-flip tests with 50,000 draws, a plus-one correction and Holm adjustment across the eight model–component contrasts. Supplementary Methods Table M4 specifies the shared random stream, test statistic and separate sensitivity analyses. Eight-mechanism cluster resampling and leave-one-mechanism-out analyses assess sensitivity to mechanism composition, with limited resolution at eight clusters. The eight model-by-component primary contrasts form the Holm family fixed for the four-model cohort before DeepSeek acquisition, after the Sol/Grok/Qwen/Mistral results had been observed. Restoration, irrelevant deletion, paraphrase, factual replacement and history compensation are secondary controls, summarized by descriptive difference intervals rather than equivalence tests. The 128-task content verification separately evaluated the prespecified rewording-versus-JSON equivalence margin of ± 0.05 .

For the supplementary specificity contrast, subtraction is performed within the same task before aggregation:

$$\hat{S}_k = \frac{1}{8} \sum_{f=1}^8 \frac{1}{n_{fk}} \sum_{e \in (f,k)} (\bar{Y}_{e,\text{irrelevant}} - \bar{Y}_{e,\text{lesion}}), \quad \bar{Y}_{eb} = \frac{1}{4} \sum_{r=1}^4 Y_{ebr}. \quad (5)$$

This supplementary contrast was defined after the primary results were inspected. It preserves task pairing and uses 5,000 task-bootstrap draws stratified within task mechanism, seed 2026092601. Two-sided task-sign-flip tests use 50,000 draws and the plus-one correction; Holm adjustment is applied to these eight supplementary contrasts as a separate family. The permutation reference assumes task-difference symmetry under its null; the bootstrap is conditional on the sampled task mechanisms. Replacement continuations are partitioned into original-only, replacement-only, overlapping targets, other valid transactions, invalid transactions and undefined replacement targets. An undefined target takes precedence in this partition and is reported rather than assigned success or failure.

The content-separation experiment uses the same preparation gate and environment evaluator. If s_e^{id} preserves the identifier subset of the agent’s memory, its principal descriptive contrast is

$$\Delta_{\text{content}|\text{id}} = \mathbb{E}_e [\Pr(Y_e(s_e) = 1) - \Pr(Y_e(s_e^{\text{id}}) = 1)]. \quad (6)$$

The saved goal and unrelated record are unchanged. Each configuration contributes 16 tasks, two per mechanism, with two continuations in each of six conditions. Paired tasks are re-sampled within mechanism for 5,000 descriptive draws (seed 2026092602). The study was designed after inspecting the whole-memory intervention and is analyzed as development evidence. All planned tasks and preparation failures are retained. These development intervals and supplementary specificity tests are reported separately from the original primary family.

The content verification used 128 new memory tasks per API configuration, with 16 tasks per mechanism and two continuations under each of eight conditions. Identity-only and content-only interventions partitioned the actual written memory. Exact JSON text and value-preserving rewording separated the effects of content, container and wording. The primary paired success differences used 10,000 task-bootstrap draws stratified by mechanism; 50,000 task-sign-flip draws and Holm adjustment evaluated the two model-specific primary contrasts. All 4,096 planned continuations entered the primary denominators. Supplementary Methods give the field partitions, secondary comparisons, encoding-equivalence analysis and complete acquisition accounting.

5.3. Predictive inference and budgets

The shared repository context is retrieved from the visible issue text using lexical path scores, with deterministic path-order tie breaking. All arms receive the same fifty candidate paths and eight source excerpts; the exact tokenization, file filters, ranking formula and excerpt limits are given in Supplementary Methods, *Shared repository retrieval*.

For the fixed linear comparison, all preprocessing is fitted inside five-fold grouped training splits (shuffled group identifiers, seed 17). Numeric features receive training-fold median imputation and standardization; model and repository use one-hot encoding with unknown categories ignored. With $\phi(x)$ denoting training-fold-transformed features, the fitted probability and incremental discrimination are

$$\hat{p}(y = 1 \mid x) = \{1 + \exp[-\beta_0 - \beta^\top \phi(x)]\}^{-1}, \quad \delta = \text{AUC}_{\text{CSB}} - \text{AUC}_{\text{generic}}. \quad (7)$$

The probability predictor is L2 logistic regression with $C = 1$, the liblinear solver, 2,000 maximum iterations, seed 17 and no class weighting. All rows for one issue remain together; visible issue near-duplicates or identical repository/base-commit pairs are additionally grouped. In this fixed linear comparison, state and generic diagnostics use the same feature count and estimator specification. The nonlinear comparison gives both arms the same representation choices, learner families and search budget; their selected representations can differ. Bridge confidence intervals repeat the grouped fit and preprocessing in 2,000 issue/template-group bootstrap draws, keeping duplicate copies of the same group within one fold. Non-estimable draws are counted. Precision-recall discrimination, Brier score, per-model results and repository- and model-held-out results complement pooled AUC. A single-class test slice is marked non-estimable. For the 100-issue reassessment, the test-cluster bootstrap conditions on predictors refitted and frozen on the four-model development cohort.

The shared three calls are charged once in joint acquisition and to each strategy in deployment accounting. Resource estimates include extra calls, transport attempts and reported input/output/reasoning tokens. Fixed-call comparisons, actual token pairs and retrospective metered-prefix analyses answer distinct budget questions. Prefix accounting includes the acquired over-budget query; unmetered responses are identified separately from measured usage.

The 100-issue reassessment conditions on a model-cohort choice made after its outcomes had been observed, with labels excluded from fitting and selection (Dwork et al., 2015). Supplementary Note 14 documents the service-driven replacement and its preceding estimates. The independent 800-issue cohort was unused in development and was acquired only after the predictors were frozen.

In addition to the fixed five-call generic diagnostic, a token-funded comparator continues generic acquisition until its reported diagnostic tokens reach or exceed those of the five CSB probes, up to ten generic calls. Token usage alone determines when generic acquisition stops. This gives the generic comparator at least as many queries and, when fully metered and successfully funded, at least as many reported tokens; the final indivisible call can overshoot. The complete results distinguish achieved token funding, incomplete funding and unmetered attempts. The primary comparison uses the token-funded generic diagnostic; the fixed-call comparison is reported separately.

For the final predictors, each diagnostic response supplies 21 reference-free quantities: confidence, validity, parsing success, path count and valid-path fraction; four agreements with the base responses; five state-answer correspondences; retrieval coverage and best/mean retrieval rank; path-query and returned-state-query lexical overlaps; serialized state length and a nonempty-state indicator. File paths are normalized, checked against the visible repository and limited to three implementation files. Set agreement is $J(A, B) = |A \cap B| / |A \cup B|$ for two nonempty sets and zero otherwise. Base-response features compare the returned set with the first base answer and the three most frequent base paths, and retain rank-weighted votes and top-path agreement. State features compare it with the input, returned and original candidate-file sets, retain its change in input-state agreement relative to the first base answer, and measure the input state’s distance from the original state. An empty input state has missing distance. Retrieval ranks range from zero to 49 among the visible candidates, with absent paths assigned 50; best and mean ranks are divided by 50 and equal one for an empty output. Lexical overlap uses lower-cased identifier tokens after stop-word removal. Supplementary Methods specify the vote normalization, tokenization and all 21 quantities.

The aggregate representation summarizes each quantity by its finite-value mean, population standard deviation, minimum, maximum and missing fraction. It additionally records pairwise file-set overlaps, the proportion of unique file sets, file-vote entropy in nats, the maximum vote count divided by the number of calls, and that call count. All funded generic calls enter these summaries. The ordered representation retains the first five response vectors, their ten pairwise overlaps and valid-response versions, and contrasts of confidence and state/base agreement with the first response. A valid-response pair is missing unless both outputs pass validation; a missing confidence is retained as missing. These additions accompany the common base features and seven original diagnostic features. Table 6 gives the selected representations; the complete ordered contrasts and column lists are specified in the Supplementary Methods and frozen model records.

Numeric features are converted to float32 before preprocessing. Within each training fold, missing values receive that fold’s median; entirely missing columns are retained and filled with zero, and indicators are added for columns containing missing training values. Numeric columns and their indicators are standardized. Model and repository are one-hot encoded, with unseen categories ignored. Every diagnostic arm had the same four representation choices and 60 configurations, evaluated in five outer issue/template-group folds and three inner folds. For each outer validation fold, inner out-of-fold AUC selected one member of each learner family. To obtain the final models, each family’s configuration was selected by its pooled five-fold out-of-fold AUC across the development cohort. Both selections broke ties by log loss and then fixed candidate order. The four probabilities are averaged with

weight 0.25 each. The selected estimators were refitted on all 300 development issues and frozen before acquiring the 800 new issues (Cawley and Talbot, 2010).

Table 6: Final configurations for the primary independent comparison. Aggregate and ordered-plus-aggregate representations contain 130 and 279 columns before preprocessing and categorical expansion. Each learner contributes one quarter of the predicted probability.

Learner	Repetition–intervention	Token-funded generic
CatBoost	Aggregate; depth 4; L2 penalty 3	Aggregate; depth 6; L2 penalty 10
Extra trees	Aggregate; depth 8; minimum leaf 5	Ordered + aggregate; depth 8; minimum leaf 5
Histogram boosting	Aggregate; maximum 15 leaves; minimum leaf 10	Aggregate; maximum 15 leaves; minimum leaf 10
Logistic regression	Aggregate; $C = 0.1$	Aggregate; $C = 0.1$

CatBoost uses 160 iterations, learning rate 0.05 and log loss. Extra trees uses 160 trees and all features at each split. Histogram boosting uses 150 iterations, learning rate 0.05, L2 penalty 5 and no early stopping. Logistic regression uses the liblinear solver with at most 3,000 iterations. Estimator random seeds are 17. The early linear comparison retains its separate $C = 1$ specification.

The primary independent comparison is repetition–intervention versus token-funded generic diagnosis. Pure state interventions and the frozen noise-contrast variant form two prespecified secondary AUC comparisons, each against its corresponding generic predictor. Ten thousand paired issue-bootstrap draws stratified by repository carry all four model records together and leave the trained estimators fixed. We report marginal 95% intervals and 98.333% intervals for the three-comparison Bonferroni family. Brier score, log loss, calibration in fixed probability bins and failure rates at fixed retained fractions are prespecified secondary descriptions. These intervals quantify new-issue variation conditional on training and the twelve sampled repositories; no new outcomes enter fitting, threshold selection, model replacement or sample-size decisions.

The generic collection starts with five calls and may continue to ten, until its recorded diagnostic tokens reach the larger of the full-state and hybrid diagnostic costs. Three common base calls are charged once in the joint acquisition and to each strategy in deployment accounting. The primary analysis retains every planned record, including terminal failures; complete-and-funded accounting is a separate sensitivity. Supplementary Notes 13–15 give the acquisition accounting for the linear comparisons.

5.4. Sampling, model-visible information and annotation

Each model in the controlled open-weight study contributes 126,000 scored rows: three scaffolds, 20 task families, 100 events per family, seven variants and three replicate indices. Of these, 108,000 are model-generation records and 18,000 are offline-policy records. The offline policy uses the frozen expected-action marginal within selected events; it is not an empirical prior fitted to structured-model outputs. The seven variants include post-hoc/scrambled control rows. The strict decisive-field lesion is a separate additional slice, not one of the seven core variants.

The seven source corpora support distinct baseline and finite-action samples. The baseline design samples up to 500 events per corpus under eight conditions and three random

seeds. The complete baseline acquisition contains 3,098 entries (3,073 unique corpus–event IDs) and all 74,352 planned responses. The finite-action sample contains 818 entries and 803 unique identifiers, from six corpora with 120 entries each and Narratives with 98. It shares 73 identifiers with the 224-event source intake. Supplementary Note 2 gives the full source-membership relationships, including the separate 497-entry snapshot, configured caps, observed counts and generation totals. The baseline replication reports raw-field coverage and reason-response contrasts separately from compatibility scores calculated using rule-completed fields.

The human evaluation sampled 1,200 items from 532,800 candidate records in the Qwen2.5-7B, 14B and 32B and Mistral-7B experiments and their controls, selecting 150 items per condition. Three graduate students in relevant disciplines, including doctoral and master’s students, independently rated every item using a common seven-dimension codebook and written instructions. The codebook specified binary judgment criteria and handling of unjudgeable cases. Annotators received the same masked items in the same randomized order and scored only visible information. Supplementary Note 5 details source-stratified sampling, masking, score aggregation and agreement analysis.

Supplementary Methods Tables M1–M3 distinguish sample roles, permitted intervention information and predictor features. Supplementary Notes 5 and 2 give the annotation procedure and sample-flow accounting, respectively.

5.5. *Reproducibility and analysis specification*

The research archive links task-level observations to model revisions, quantization, inference settings, complete prompts, task-visible responses, environment snapshots and analysis inputs. Versioned specifications record cohort membership, interventions and scoring rules. Prespecified tests and exploratory analyses have their respective data roles and comparison families defined in the statistical methods. Source hashes and request records connect the full planned samples to the reported summaries.

6. Discussion

Agent-written state contributes to action through the task information it carries. The controlled experiments establish consistent responses to supplied information; the execution experiments identify dependence on records produced and retained by the acting system. In Sol and DeepSeek, removing memory or a persistent goal reduces transaction success, restoration recovers performance, and irrelevant perturbations have much smaller effects. Preserving transaction identifiers while removing other memory content retains a substantial deletion effect across all eight mechanism-specific point estimates. The evidence connects a realized record to executable choices and shows which parts of that record matter.

The information-access controls explain why the same component can matter strongly in one context and little in another. Relevant facts restated at handoff reduce reliance on saved memory, and access to earlier history compensates for its removal. Dependence therefore reflects the available routes by which task information reaches execution. This finding complements the distinction between retrieval and utilization in memory research (Shen et al., 2026; Yuan et al., 2026): the fixed checkpoint holds writing constant, while restoration and alternative access vary what the continuing agent can use. A small deletion

effect in a redundant context can coexist with effective use of the same information through another source.

Content replacement gives a directional interpretation to this dependence. Actions move toward the target specified by the altered record, even when that target departs from the original task requirement. This separates state use, record correctness and successful completion. The crossed target scores extend the intervention logic used to study reasoning and causal representations to parameterized transactions (Paul et al., 2024; Geiger et al., 2021). They also suggest a practical distinction in diagnosing memory failures: an agent may lack the relevant information, fail to use accessible information, or act consistently on an inaccurate record. Deletion, compensation and replacement help distinguish these cases within the tested task design.

Encoding also affects how information is used. The content experiment compares a structured record, its exact JSON serialization and value-preserving rewording, so the effect of removing facts can be interpreted alongside changes in presentation. Sol’s difference between the two text conditions shows that preserved field values alone do not imply identical execution probabilities. DeepSeek’s descriptive equivalence result applies to the stated margin and task sample. This model dependence accords with intervention studies in which reliance on intermediate text varies across tasks and configurations (Lanham et al., 2023). Output agreement and human ratings characterize the reported material, while executable outcomes establish its connection to action at the exposed state interface.

The software application evaluates a further consequence: whether response relations help predict localization success. The repetition–intervention diagnostic produced an independent AUC point increment of 0.00382 and a Brier-score reduction of 0.00293 relative to token-funded generic queries. The marginal AUC interval was positive, while the registered family interval included zero. The result supports a modest pooled predictive signal for the combined diagnostic, with family-level superiority unresolved and gains varying by model. Agreement and uncertainty across repeated generations already contain useful information (Manakul et al., 2023; Farquhar et al., 2024); the matched comparator measures the increment over that information. Behavioral dependence and predictive discrimination consequently answer distinct questions: a state can influence actions strongly even when its intervention responses add little to an already informative predictor.

The experimental scope is defined by eight synthetic computational mechanisms, short tool horizons and a shared JSON interface. Four continuations per core condition and two per content-verification condition give discrete task-level probabilities. The content verification includes 16 new tasks per mechanism and separates remembered facts from model-dependent encoding effects. Quantized open-weight performance near the floor restricts inference about state dependence for those configurations. The independent prediction comparison concerns unused issues from twelve repositories and fixed trained predictors; output agreement concerns visible reports at the stated scoring coverage. Longer interactions would test how these relationships persist as records accumulate, conflict or become obsolete. Equal-information comparisons of repair policies would address the separate question of how diagnosing a consequential record can improve task completion.

7. Conclusion

Structured state supports consistent controlled action responses, and agent-written memories and goals make substantial, recoverable contributions to tool execution in the tested API configurations. Specificity, restoration and alternative information access show when a saved record matters. Content separation and replacement distinguish the use of task information from the need for identifiers and from the correctness of the resulting action. On new software issues, the hybrid diagnostic yielded modestly higher pooled discrimination and lower Brier loss than token-funded generic queries, although the AUC difference did not meet the prespecified three-comparison family criterion. Together, the experiments connect the information an agent retains to the actions it subsequently executes, providing an empirical account of state-dependent action.

Data and code availability

Analysis-ready data and code supporting the findings are publicly available at <https://github.com/JiaXiao-Peter/Causal-State-Binding>. The repository provides study protocols, cohort membership lists, task-level outcomes, frozen localization predictions and estimators, scoring and analysis code, and source data for all figures. The versioned research package includes commands to reproduce the reported summaries and figures from saved observations. Third-party benchmark materials retain their original licenses and access terms.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

ChatGPT and Codex (OpenAI) assisted with code revision and grammatical editing. Responsibility for the interpretation, scientific claims and final text rests with the author.

CRedit authorship contribution statement

Xiao Jia: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review and editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The author thanks the maintainers of the public datasets and open methodological resources used in this study.

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